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Beyond Linear Reasoning: Combining Chain-of-Thought with A* Search for Robust Multi-Strategy Inference

 ICLR

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Summary

This paper proposes a budgeted A*-style search procedure over natural-language reasoning traces as a complementary alternative to linear chain-of-thought (CoT) decoding in LLMs. It defines a path cost over step length and self-reported step confidence, introduces a conservative depth-plus-coverage heuristic with an admissibility guarantee under explicit assumptions, and empirically studies complementarities between CoT and A* across StrategyQA, LogicQA, and HotpotQA. The central empirical finding is that CoT and A* solve different subsets of questions, yielding large oracle gains; simple routing strategies (semantic entropy, a feature-engineered random forest, and an LLM-as-critic) recover part of this gap but leave significant headroom.

Strengths

Technical novelty and innovation

- The paper formalizes A* search over natural-language reasoning states with an explicit path-cost objective and a conservative heuristic, accompanied by a simple but clear admissibility statement under stated assumptions. While not claiming end-to-end optimality, the formulation helps ground search-time decisions.
- The work focuses on complementarity between linear and search-based reasoning and frames the problem as routing, which is a useful reframing compared to “replace CoT with search.”

Experimental rigor and validation

- The instance-level complementarity analysis (including an oracle and the four-way decomposition: both-correct, CoT-only, A*-only, both-wrong) is informative and consistently shows sizable oracle gaps.
- The paper evaluates multiple routers and includes a qualitative taxonomy of regimes where each strategy succeeds or fails, plus small-scale human-style annotations that link correctness to coverage rather than stylistic fluency.

Clarity of presentation

- The motivation is well-articulated, and the qualitative analyses are concrete and insightful (e.g., non-linear aggregation, hypothesis elimination, multi-hop chaining vs. direct factual recall and globally coherent argumentation).
- Assumptions and scope of the theorem are clearly caveated (admissibility for the underlying path-cost ordering, not for the budgeted voting procedure).

Significance of contributions

- The empirical message—that search and linear decoding are complementary and the main challenge is routing—is timely and relevant for the community’s ongoing effort to make LLM reasoning more reliable.
- The results suggest a research direction toward adaptive inference-time strategy selection, potentially spawning work on better routers



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Tech Overview



Weaknesses

Technical limitations or concerns

- The heuristic's coverage term $E(n)$ is underspecified. It depends on "target entities," but the paper does not describe how targets are defined per dataset, how they are extracted at inference time, whether supervision beyond the question/context is used, or how leakage is prevented (especially for HotpotQA).
- The step confidence $c^{(j)}$ is "self-reported" by the same model used for generation. Without a robust calibration procedure and ablations, the path cost can be gamed or be poorly correlated with step quality. Bounding $c \in [0.5, 0.99]$ is ad hoc; details on elicitation, calibration, and sensitivity are missing.
- The admissibility theorem is contingent on $\omega_d + \omega_c \leq c_{\min}$, which, for high $c_{\max_conf}$, effectively forces very small heuristic weights. The practical utility of this guarantee is unclear, since the actual inference is budgeted and uses multi-goal voting; there is no evidence that the admissibility constraint meaningfully influences or improves empirical search.

Experimental gaps or methodological issues

- Missing compute and budget reporting: no clear accounting of node budgets B , depth limits $D_{\max}$, N_c , token costs, wall-clock, or fairness controls relative to Self-Consistency, Tree-of-Thoughts, and other structured search baselines. Without cost/efficiency analysis, it is hard to judge practical trade-offs.
- Several critical implementation details are missing or ambiguous, e.g., how multi-goal voting aggregates answers (Eq. (??) placeholder), how $H_{\min}$ is set per dataset, and whether HotpotQA is open- or closed-book (and how contexts are provided).
- No ablations for the heuristic components (depth vs. coverage vs. none) and for confidence-weighted path cost vs. pure length. It remains unclear which ingredients actually drive the observed gains.
- Limited scale: only one main backbone (LLaMA-3.1-8B) and one critic (DeepSeek-R1-Distill-Qwen-14B) are considered; stronger models or diverse families would help generality claims.

Clarity or presentation issues

- Table 1 contains an apparent anomaly: "LLM-as-Judge Router" shows 73.45% on LogicQA (vastly out of line with all other LogicQA numbers and likely a typo). Also, "LLM-as-Judge" vs. "LLM-as-Critic" naming is confusing; please standardize and clarify.

- Algorithm 1 returns a “weighted vote over GOALS using Eq. (??)”,
- indicating a missing equation reference. Some equations have minor formatting artifacts.

Missing related work or comparisons

- The **paper cites ToT and GoT** but does not compare to or discuss more recent search/routing/steering methods that target similar goals:
 - NCoTS (operator-level lookahead search at delimiter tokens), which pursues lightweight search-time control with empirically favorable cost–accuracy trade-offs.
 - RouteGoT (node-adaptive routing under global budget constraints inside a GoT graph), which directly tackles budgeted structured reasoning with learned routing policies.
 - Activation steering/routing methods (e.g., RISER), exploring adaptive reasoning without external search.
 - RL-based shaping for reasoning efficiency (e.g., T2T) and cache-based approaches (ReasonCACHE) that learn reusable reasoning trajectories without parameter updates.
- Without positioning against these lines, the paper’s novelty and empirical value are harder to assess.

Detailed Comments

Technical soundness evaluation

- The formalization of A* over natural-language states is sensible, but the heuristic’s practical definition is underspecified. In particular:
 - The **coverage metric $E(n)$ could leak supervision if it depends on oracle entities; please clarify whether it is computed only from question/context text via unsupervised NER/linking, or if it uses target annotations (e.g., Hotpot supporting facts).**
 - Confidence $c^{(j)}$ requires careful calibration; otherwise, the search priority ($\text{length} + 1 - c$) may not correlate with correctness. Please provide calibration info and sensitivity to c-quality (e.g., prompt variants, temperature, or replacing self-reports with log-prob proxies).

- The admissibility guarantee is mathematically correct under its assumptions, but its practical implications are limited by (a) the budgeted, multi-goal nature of the actual algorithm, and (b) the very tight bound on $\omega_d + \omega_c$ implied by $c_{\min}$. Without ablations showing that enforcing that constraint meaningfully helps, this reads as a reassuring but largely decorative result.

Experimental evaluation assessment

- The complementarity analysis is a strength: oracle gaps are consistently large, and disagreement-set breakdowns are compelling.
- However, the paper omits compute/cost metrics. Given that routing and search aim to improve cost-accuracy trade-offs, reporting tokens, expansions, and latency is crucial. Several recent works emphasize efficiency; a direct comparison would strengthen the case.
- **Baseline fairness needs elaboration: how were ToT and Self-Consistency hyperparameters chosen?** Are the search budgets matched by comparable token limits? Please report hyperparameters for all methods clearly and align costs where possible.
- Missing ablations: heuristic components, confidence- vs. length-based costs, and effect of $H_{\min}$ choice. Also analyze how different node budgets B and candidate counts N_c affect both accuracy and compute.
- Statistical reliability: provide confidence intervals or multiple runs (especially for routers trained with features) to quantify variance.

Comparison with related work (using the summaries provided)

- **NCoTS (Neural Chain-of-Thought Search) offers a light-touch operator-level lookahead and learned scoring to improve reasoning efficiency. It is a direct point of comparison for "search-time control of reasoning" with explicit cost considerations, in contrast to full tree search. A discussion and empirical comparison would contextualize the proposed A* approach.**
- RouteGoT introduces budget-aware routing within a graph-of-thoughts and demonstrates strong cost-accuracy improvements with learned policies. Since this paper's key message is "routing matters," RouteGoT is especially relevant; a conceptual and empirical comparison would be valuable.

- RISER focuses on activation-space routing/steering to adaptively enhance reasoning without external search. It bears on the “routing across strategies” narrative, albeit at the latent level. Briefly discussing similarities and differences would broaden the context.
- T2T shows that training-time reward shaping can simultaneously encourage exploration on hard items and compression on easy ones—aligned with the paper’s observation that not all queries benefit from search. Mentioning such complementary training-time techniques could help readers navigate design choices.
- ReasonCACHE indicates a middle path between ICL and in-weight learning through KV-cache distillation; relevant to “efficient reuse of reasoning strategies,” especially if one hopes to amortize search over time.
- The survey on reasoning failures (2602.06176) reinforces the significance of the problem; connecting identified failure modes (e.g., brittle composition, robustness issues) to the paper’s observed regimes would strengthen the broader impact narrative.

Discussion of broader impact and significance

- The central insight—that questions differ in “reasoning topology,” and routing should match strategy to problem structure—is important and has practical implications for real systems that need to balance reliability and cost.
- However, realizing this vision requires stronger, better-calibrated signals (confidence, coverage) and more principled routing. The paper convincingly surfaces the opportunity but still falls short of demonstrating a robust, general routing solution.

Questions for Authors

1. How exactly is $E(n)$ (coverage) computed on each dataset? What are “target entities,” how are they extracted, and can you rule out label leakage (e.g., using Hotpot supporting facts at test time)? Please provide algorithmic details and an ablation removing $E(n)$.
2. How are self-reported confidences $c^{(j)}$ elicited and calibrated? Do models tend to inflate c ? Please include sensitivity analyses (e.g., using

token log-probs as proxies, disabling c , or temperature variations) and show how accuracy varies with c -quality.

3. What are the exact budgets (B , $D_{\max}$, N_c), prompt templates, sampling temperatures, and token limits for all methods (A^* , CoT, Self-Consistency, ToT), and how do their compute costs compare (tokens, latency)? Can you match or normalize compute for a fairer comparison?
4. Please clarify the “multi-goal voting” rule (Eq. (??) is missing). How are goal nodes weighted? Does this depend on costs, confidences, or heuristic values?
5. The LogicQA number for “LLM-as-Judge Router” in Table 1 (73.45%) appears inconsistent with other results. Is this a typo? If not, please explain the discrepancy and provide variance estimates.
6. How is $H_{\min}$ chosen per dataset? Is it fixed, learned, or estimated on the fly? How sensitive are results to $H_{\min}$ and to the admissibility constraint $\omega_d + \omega_c \leq c_{\min}$?
7. Could you compare to at least one recent budget-aware structured reasoning method (e.g., NCoTS, RouteGoT) and report efficiency metrics? If not feasible, please discuss differences in assumptions and expected trade-offs.

Overall Assessment

Estimated Score: 5.2/10 (Calibrated to ICLR scale)

This paper makes a useful empirical and conceptual contribution by showing that linear CoT and search-based reasoning are substantially complementary, and by framing the key challenge as routing rather than wholesale replacement. The qualitative analyses and oracle-gap decompositions are strong and support the thesis that reasoning “topology” varies by instance. However, the technical novelty is moderate, the admissibility result has limited practical bite given the budgeted/voting setup, and several important methodological details are missing (coverage definition, confidence calibration, multi-goal voting, cost/budget reporting). The evaluation would be substantially stronger with compute-normalized comparisons, ablations of the heuristic and confidence signals, and positioning against more recent search/routing baselines such as NCoTS

and RouteGoT. I view the paper as promising and thought-provoking, but not yet at the level of rigor and completeness expected for ICLR. With clarifications, ablations, and cost-aware baselines, it could become a solid contribution.

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